LECTURE NOTES

Computer Networks

Network Devices and Physical Structures

- Hardware Components
- Software Components
- IP Addressing
- Connection Types
- Physical Topologies

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2 hrs Theory

Where We Left Off in Week 1

Last week we said "nodes and links". This week we open both of them up.

> *"A computer network is a collection of two or more autonomous computing devices, interconnected by a transmission medium, that exchange data according to a shared set of rules."*

Week 1 said "Nodes"

But which devices, exactly? And what does each one actually do to a packet?

→ **Part 1 today**

Week 1 said "Protocols run on them"

But what software makes a card, a driver, and a protocol stack work together?

→ **Part 2 today**

Week 1 said "Every node has an address"

But what is an IP address made of, and how do you read or assign one?

→ **Part 3 today**

Week 1 said "Links join them"

But in what shape? A line, a star, a ring? The choice changes everything.

→ **Parts 4 and 5 today**

Session Roadmap

Five theory parts, then the lab

01

Hardware Components

NIC, hub, switch, router, modem, WAP

02

Software Components

Drivers, stacks, OS services

03

IP Addressing

How devices get identified

04

Types of Connection

Point-to-point and multipoint

05

Physical Topology

Bus, star, ring, mesh, tree, hybrid

06

Practical

Topologies + addressing in the lab

LESSON

Weekly Learning Outcomes

By the end of this session you will be able to:

01

Identify Networking Hardware

Name each common device and state exactly what it does to traffic passing through it.

02

Distinguish Connection Types

Tell a point-to-point link from a multipoint link, and say when each is used.

03

Compare Physical Topologies

Weigh bus, star, ring, mesh, tree, and hybrid on cost, fault tolerance, and scalability.

04

Read and Assign IP Addresses

Interpret an IPv4 address and mask, and configure a device correctly.

01

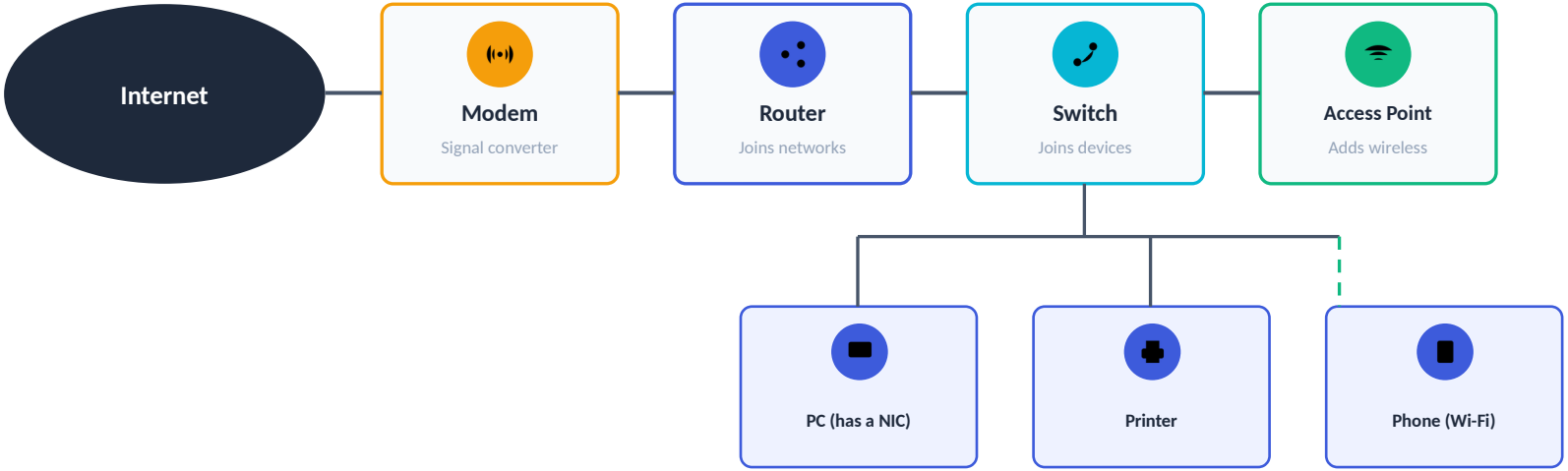
Hardware Components

The physical devices that make a network work, and what each one really does

NIC | Repeater | Hub | Switch | Router | Modem | Access Point

The Big Picture: Where Each Device Sits

One typical home or small office, with every device we will study



Read it left to right: the signal arrives from outside, gets converted, gets routed, gets distributed, and finally reaches a device that has a NIC inside it.

Network Interface Card (NIC)

The part that makes a computer capable of networking at all

A NIC is the hardware inside every networked device that converts the computer's internal data into signals for the cable or the air, and back again. No NIC means no network connection, no matter how many cables you plug in.

How it works

1. Takes data from the operating system in the computer's own format.
2. Packages it into frames and adds the source and destination MAC addresses.
3. Converts the frame into electrical, light, or radio signals for the medium.
4. Listens continuously and reverses the whole process for incoming signals.

Key facts

OSI layer	Layer 1 and 2 (Physical + Data Link)
Also called	Network adapter, LAN card, Ethernet card
Unique ID	Burned-in MAC address, 48 bits
Types	Wired (RJ45), wireless, fiber, USB
Speed	Typically 100 Mbps, 1 Gbps, or 10 Gbps

Check it yourself: run `ipconfig /all` on Windows or `ip link` on Linux. The Physical Address you see is your NIC's MAC address.

Repeater

The simplest device on the network: it only fights distance

Every signal weakens as it travels down a cable. This is called attenuation. A repeater sits in the middle of a long run, receives the weakened signal, regenerates a clean copy at full strength, and sends it onward.

How it works

1. Receives an incoming signal that has become weak and distorted.
2. Regenerates it — it rebuilds the signal, it does not merely amplify noise.
3. Retransmits the clean signal at full strength out of the other port.
4. Makes no decisions: it never looks at addresses or drops anything.

Key facts

OSI layer	Layer 1 (Physical) only
Ports	Typically 2
Reads addresses?	No — it has no idea what data is
Problem solved	Attenuation over distance
Modern use	Wi-Fi range extenders, fiber links

Why it matters: Ethernet over copper is limited to about 100 metres. A repeater is how you cross that limit without changing the cable type.

Hub

A multi-port repeater — and a cautionary tale

A hub connects several devices together, but it has no intelligence at all. When a frame arrives on one port, the hub copies it out of every other port. Every device receives every frame, checks the destination address, and discards what is not for it.

How it works

1. A frame arrives on one of the hub's ports.
2. The hub floods an identical copy out of every other port.
3. Every connected device receives it and inspects the destination address.
4. One device keeps it; all the others waste effort discarding it.

Key facts

OSI layer	Layer 1 (Physical)
Forwarding	Floods to all ports, always
Collision domain	One shared domain for everyone
Bandwidth	Shared — 100 Mbps split among all
Status today	Obsolete; replaced by switches

The fatal flaw: if two devices transmit at once, the signals collide and both must retry. The more devices you add, the worse it gets. Hubs do not scale.

Switch

The device that actually runs every modern LAN

A switch looks like a hub but is fundamentally smarter. It learns which device is on which port by reading MAC addresses, stores that in a table, and then forwards each frame only to the port where the destination actually is.

How it works

1. A frame arrives. The switch reads its source MAC address.
2. It records “this MAC lives on this port” in its MAC address table.
3. It reads the destination MAC and looks it up in that same table.
4. It forwards the frame out of that one port only. Everyone else is undisturbed.

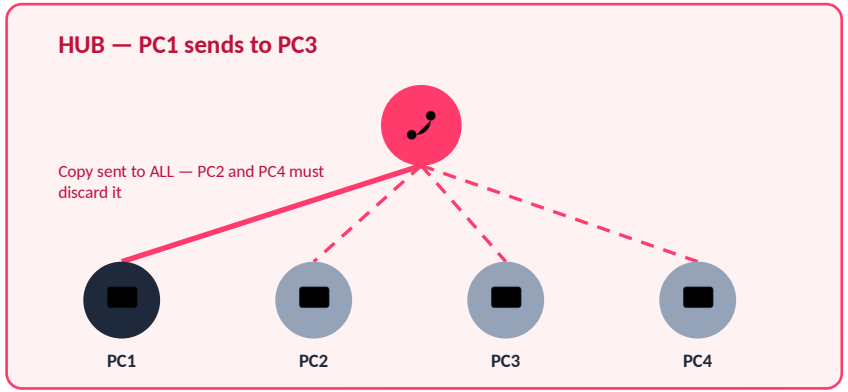
Key facts

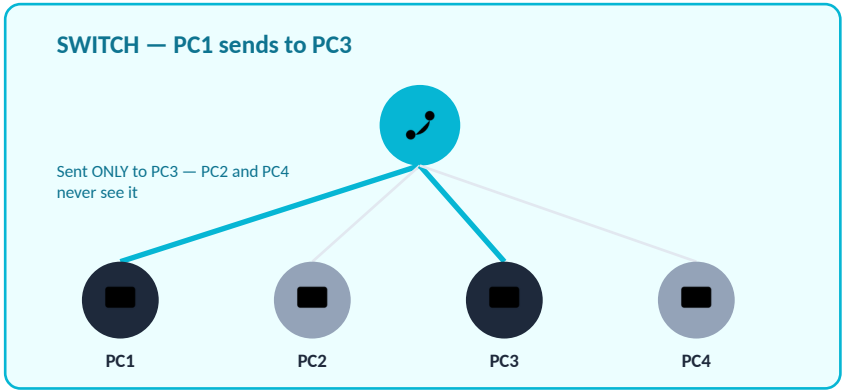
OSI layer	Layer 2 (Data Link)
Forwarding	Only to the correct port
Collision domain	One per port — so collisions vanish
Bandwidth	Full speed on every port, at once
Learns	MAC address table, built automatically

First frame exception: if a destination MAC is not yet in the table, the switch floods that one frame like a hub, then learns from the reply and never floods it again.

Hub vs Switch: Watch the Difference

Same cable layout, completely different behaviour





Feature	Hub	Switch
Forwarding decision	None — floods everything	Reads MAC, forwards to one port
OSI layer	Layer 1 (Physical)	Layer 2 (Data Link)
Collision domains	One, shared by all ports	One per port
Effective bandwidth	Shared between all devices	Full speed per port
Security	Anyone can capture all traffic	Traffic is not exposed to others

Router

The device that joins one network to another

A switch moves frames inside a single network. A router moves packets between different networks. It reads IP addresses, consults a routing table, and chooses the best next hop toward the destination — possibly on the other side of the world.

How it works

1. A packet arrives. The router reads its destination IP address.
2. It compares that IP against every entry in its routing table.
3. It picks the most specific matching route and its next-hop interface.
4. It rewrites the frame header, decrements TTL, and forwards the packet.

Key facts

OSI layer	Layer 3 (Network)
Reads	IP addresses, not MAC addresses
Separates	Broadcast domains — one per interface
Builds	Routing table, static or via RIP/OSPF
Extras	Usually also does NAT, DHCP, firewall

The one-line difference: a switch asks “which port is this device on?” A router asks “which network is this address in, and how do I get closer to it?”

Modem

MOdulator + DEModulator — a translator between two kinds of signal

Your computer speaks in digital bits. The line coming into the building may carry analogue signals, cable TV frequencies, or optical light. A modem converts between the two so that digital data can survive the journey across that medium.

How it works

1. Outgoing: takes digital data from the router and modulates it onto the carrier.
2. The modulated signal travels over the telephone line, coaxial cable, or fiber.
3. Incoming: demodulates the arriving carrier signal back into digital bits.
4. Hands the clean digital data to the router for onward delivery.

Key facts

OSI layer	Layer 1 (Physical)
Job	Signal conversion, not routing
Types	DSL, cable, fiber (ONT), 4G/5G dongle
Connects	Your premises to the ISP
Common today	Combined modem + router in one box

Point of confusion: the box your ISP gives you is usually a modem, router, switch, and access point in one case. Four devices, four jobs, one plastic shell.

Wireless Access Point (WAP)

The bridge between the wired network and the air

An access point plugs into the wired network with a cable and offers a wireless network to devices around it. It is essentially a switch for radio: it takes Wi-Fi frames from the air, puts them onto the cable, and does the reverse for traffic coming back.

How it works

1. Broadcasts a network name (SSID) that nearby devices can discover.
2. Authenticates a joining device, usually with a WPA2 or WPA3 password.
3. Converts wireless 802.11 frames into wired Ethernet frames, and back.
4. Coordinates airtime so overlapping devices do not talk over each other.

Key facts

OSI layer	Layer 2 (Data Link)
Standard	IEEE 802.11 (Wi-Fi 5, 6, 6E, 7)
Bands	2.4 GHz (range) and 5 GHz (speed)
Medium	Shared — like a hub, devices take turns
Scaling up	Many APs, one controller, one SSID

Worth remembering: wireless is a shared medium. Twenty phones on one access point share its capacity, much as twenty PCs once shared a hub.

All Seven Devices, Side by Side

If you learn one table from Week 2, learn this one

Device	Layer	What it does	Reads	Where you find it
NIC	1–2	Connects a device to the medium	Frames + MAC	Every networked device
Repeater	1	Regenerates a weakened signal	Nothing	Long cable runs
Hub	1	Floods frames to every port	Nothing	Obsolete
Switch	2	Forwards frames to the right port	MAC address	Every modern LAN
Router	3	Moves packets between networks	IP address	Joining LAN to WAN
Modem	1	Converts digital to line signal	Nothing	The ISP connection
Access Point	2	Extends the LAN over radio	MAC address	Wireless coverage

The pattern: the higher the OSI layer a device works at, the more it understands about the traffic — and the smarter its decisions can be.

02

Software Components

Hardware moves the signal. Software decides what the signal means.

- Drivers
- Protocol stacks
- Network OS services

The Software Side of a Network

Three layers of software sit between your application and the cable

Your Application	A browser, an email client, a game. It just asks for data by name.
▼	
Network OS Services	File sharing, printing, user accounts, DHCP, DNS — the services the network offers.
▼	
Protocol Stack	TCP/IP. Adds addressing, breaks data into packets, guarantees delivery.
▼	
Device Driver	Translates the OS's generic requests into commands this exact NIC understands.
▼	
NIC Hardware	Turns bits into electrical, optical, or radio signals on the medium.

Read it downward: each layer only talks to the one directly below it. That is why you can change your Wi-Fi card without rewriting your browser.

Device Drivers

The small piece of software that lets an operating system speak to one specific card

There are thousands of different NIC models, each with its own registers and quirks. An operating system cannot know them all. Instead it issues one generic instruction — “send this frame” — and the driver, written by the card's manufacturer, turns that into the exact commands this card needs.

It is an abstraction layer

The OS writes code once. Every NIC vendor writes a driver to match that interface.

It exposes settings

Link speed, duplex mode, MTU, and power management are all driver-controlled.

It needs updating

Driver updates fix bugs, add security patches, and often improve throughput.

It is a common fault

A card that shows in Device Manager with a warning is usually a driver problem, not a hardware failure.

Troubleshooting rule: if a machine has link lights but no connectivity at all, check the driver before you replace the card.

Network Operating System Services

The features an operating system offers once it is connected to others

Every modern OS — Windows Server, Linux, macOS — includes networking services. These are the things that make a collection of connected machines behave like one managed system rather than a set of strangers on the same cable.

File Sharing

Shared folders and network drives, with permissions.

SMB, NFS

Print Services

One printer queue serving many users, with spooling.

CUPS, Windows Print

Authentication

One username and password that works on every machine.

Active Directory, LDAP

DHCP

Hands out IP addresses automatically to devices that join.

Built into most routers

DNS

Resolves names into IP addresses so people use words.

BIND, Windows DNS

Security & Access Control

Firewall rules, user rights, and audit logging.

iptables, Group Policy

Two of these matter today: DHCP hands out addresses and DNS resolves names — both are about to become very relevant in the IP addressing section.

03

IP Addressing

How every device on a network gets a unique, meaningful identity

MAC vs IP | IPv4 structure | Binary | Subnet masks | Classes | Private & public | DHCP | Subnetting | IPv6

Why Do We Need Addresses At All?

A switch has a hundred ports. How does a packet know which one leads home?

Consider posting a letter. Your name identifies you personally and never changes. Your street address identifies where you currently are, and changes when you move. A network needs both kinds of identity, for exactly the same reasons — and it has both.

	MAC Address (physical)	IP Address (logical)
What it is	A permanent hardware serial number	A logical, assignable location
Set by	The manufacturer, burned into the NIC	The network administrator or DHCP
Format	48 bits, hex: 00:1A:2B:3C:4D:5E	32 bits, decimal: 192.168.1.10
Changes when you move?	No — it travels with the device	Yes — a new network means a new IP
Works across routers?	No — local segment only	Yes — this is what makes the Internet work
OSI layer, used by	Layer 2 (Data Link) — switches	Layer 3 (Network) — routers

Both are needed: IP gets the packet to the right network anywhere on Earth. MAC then gets it to the right device on that final cable.

The Structure of an IPv4 Address

32 bits, written as four decimal numbers separated by dots

192	.	168	.	1	.	10
11000000		10101000		00000001		00001010
Octet 1 · 8 bits		Octet 2 · 8 bits		Octet 3 · 8 bits		Octet 4 · 8 bits

32 bits total

Four octets of 8 bits each. 8 × 4 = 32 bits.

Range 0 to 255

Eight bits can hold 2^8 = 256 values, counted from 0.

About 4.3 billion addresses

2^{32} total. It sounded infinite in 1981. It is not.

Written in dotted decimal

Humans read 192.168.1.10; the machine only ever sees the bits.

Why 255 and not 256? because counting starts at 0. The eight bits 11111111 equal 255, and 00000000 equals 0 — that is 256 different values.

Reading an Octet in Binary

You cannot subnet without this. It takes five minutes to learn.

Each of the eight bits in an octet has a fixed place value. To convert, add up the place values wherever there is a 1.

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0
+128	+64	0	0	0	0	0	0

11000000 = 128 + 64 = 192

Practise these

255	192	168	10	1
11111111	11000000	10101000	00001010	00000001
128+64+32+16+8+4+2+1	128+64	128+32+8	8+2	1

Memorise the eight place values — 128, 64, 32, 16, 8, 4, 2, 1. Every subnetting question you will ever face reduces to adding some of those numbers.

Network Portion and Host Portion

An IP address is two addresses in one: which network, and which device on it

A telephone number splits into an area code and a subscriber number. An IP address splits the same way: the left part identifies the network, the right part identifies one device inside it. The subnet mask is what tells you where the split falls.

192 . 168 . 1	. 10
NETWORK PORTION — which network	HOST PORTION — which device
255 . 255 . 255	. 0

Subnet mask 255.255.255.0 — a 255 means "this octet is network", a 0 means "this octet is host"

Same network portion

Two devices can talk directly, through a switch. No router needed.

Different network portion

Traffic must go through a router, using the default gateway.

CIDR shorthand

255.255.255.0 is written /24 — meaning 24 network bits.

The Week 1 lab, explained: PC0 was 192.168.1.10 and PC1 was 192.168.1.11, both /24. Same network portion, so ping worked with no router at all.

IPv4 Address Classes

The original scheme from 1981 — still worth knowing, because the vocabulary survives

Class	1st octet	Default mask	Networks	Hosts per network	Typical use
A	1 – 126	255.0.0.0 (/8)	126 networks	16,777,214 hosts each	Very large organisations
B	128 – 191	255.255.0.0 (/16)	16,384 networks	65,534 hosts each	Medium organisations
C	192 – 223	255.255.255.0 (/24)	2,097,152 networks	254 hosts each	Small networks, homes
D	224 – 239	Not applicable	Multicast	One-to-many delivery	Streaming, routing updates
E	240 – 255	Not applicable	Reserved	Experimental use only	Never assigned

127 is missing on purpose

The whole 127.x.x.x block is reserved for loopback — that is why Class A stops at 126.

Classes are obsolete in practice

Since 1993 we use CIDR, which lets the split fall anywhere. But the names persist in exams and conversation.

Private and Public Addresses

Why your laptop and millions of others all have the address 192.168.1.10

IPv4 ran out of addresses. The solution was to reserve three blocks that anyone may reuse freely inside their own network, on the condition that those addresses never appear on the public Internet. A router translates them on the way out.

The three private ranges (RFC 1918)

10.0.0.0 – 10.255.255.255	10.0.0.0/8	16.7 million addresses	Large enterprises
172.16.0.0 – 172.31.255.255	172.16.0.0/12	1 million addresses	Medium networks
192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536 addresses	Homes, small offices, labs

NAT: how private addresses reach the public Internet



Addresses You Cannot Assign

In every network, some addresses are already spoken for

192.168.1.0 — **Network address**

All host bits are 0. It names the network itself, never a device.

192.168.1.255 — **Broadcast address**

All host bits are 1. A packet sent here goes to every device on that network.

127.0.0.1 — **Loopback**

Refers to this machine itself. Pinging it tests your own TCP/IP stack.

169.254.x.x — **APIPA / link-local**

A device assigns this to itself when DHCP fails. It signals a problem.

0.0.0.0 — **Unspecified / default**

Means “any address” or, in a routing table, “everywhere else”.

255.255.255.255 — **Limited broadcast**

Reaches every device on the local segment. Routers never forward it.

The practical consequence: a /24 network has 256 addresses but only 254 usable hosts — the first is the network address and the last is the broadcast.

How a Device Gets Its Address

Two ways: you type it in, or the network hands it over

Static Addressing

An administrator types the address into the device by hand.

Good for	Servers, printers, routers, switches
Advantage	The address never changes, so it can be relied upon
Disadvantage	Slow to configure, easy to duplicate by mistake
Failure mode	Two devices with the same IP — both break

Dynamic Addressing (DHCP)

A DHCP server leases an address automatically on join.

Good for	Laptops, phones, guest devices, labs
Advantage	Zero configuration, no duplicates possible
Disadvantage	The address may change between sessions
Failure mode	No DHCP server reachable → device self-assigns 169.254.x.x

The DHCP handshake: DORA

D — Discover

Client broadcasts: is there a DHCP server out there?

O — Offer

Server replies with an available address it could lease.

R — Request

Client formally asks for that specific offered address.

A — Acknowledge

Server confirms the lease and its duration. Done.

Subnetting: Splitting One Network Into Several

Borrow bits from the host portion and give them to the network portion

A university gets one address block. It does not want every department, lab, and office on one flat network — that would be a single huge broadcast domain with no separation. Subnetting divides that one block into several smaller, independent networks.

Security

Student and staff traffic can be separated and filtered between subnets.

Less broadcast noise

A broadcast stays inside its own subnet instead of reaching everyone.

Organisation

One subnet per building or department makes the network legible.

Performance

Smaller collision and broadcast domains mean less contention.

CIDR notation · usable hosts = $2^n - 2$, where n is the number of host bits left

CIDR	Subnet mask	Total addresses	Usable hosts	Typical use
/24	255.255.255.0	256	254	A standard small LAN
/25	255.255.255.128	128	126	Half a /24 — two subnets
/26	255.255.255.192	64	62	A quarter — four subnets
/27	255.255.255.224	32	30	Eight subnets
/30	255.255.255.252	4	2	A link between two routers

Worked Example: Subnetting 192.168.1.0/24

Split one Class C network into four equal subnets

Task: the Department of Computing has 192.168.1.0/24 and needs four separate subnets — one each for Staff, Lab A, Lab B, and Wi-Fi.

1	**How many subnets do we need?**	Four. We need 2 borrowed bits, because $2^2 = 4$.
2	**Borrow those bits from the host portion**	/24 becomes /26. 24 + 2 = 26 network bits.
3	**Work out the new mask**	Two bits into the last octet: 128 + 64 = 192, so 255.255.255.192.
4	**Work out the block size**	256 − 192 = 64. Each subnet covers 64 addresses.
5	**Count the usable hosts**	6 host bits remain: $2^6 - 2 = 62$ usable hosts per subnet.

Subnet	Network address	Usable host range	Broadcast address
Staff	192.168.1.0/26	192.168.1.1 – .62	192.168.1.63
Lab A	192.168.1.64/26	192.168.1.65 – .126	192.168.1.127
Lab B	192.168.1.128/26	192.168.1.129 – .190	192.168.1.191
Wi-Fi	192.168.1.192/26	192.168.1.193 – .254	192.168.1.255

IPv6: The Long-Term Answer

IPv4 gave us 4.3 billion addresses. We needed rather more than that.

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Feature	IPv4	IPv6
Address length	32 bits	128 bits
Total addresses	About 4.3 billion	About 340 undecillion (3.4×10^{38})
Notation	Dotted decimal: 192.168.1.10	Hexadecimal, colon-separated
Configuration	Manual or DHCP	Can self-configure (SLAAC)
NAT required?	Yes, in practice always	No — every device can have a public address
Broadcast	Yes	No — replaced by multicast and anycast
Header	Variable length, more complex	Fixed 40 bytes, simpler to process

Where we are now: both run side by side, a situation called dual stack. IPv4 will be with us for years yet, so you must know both.

Commands You Will Use Constantly

Five commands that answer almost every basic addressing question

`ipconfig /all`	Windows	Shows your IP address, subnet mask, default gateway, DNS servers, and MAC address.
`ifconfig / ip addr`	Linux, macOS	The same information on Unix-like systems. ip addr is the modern form.
`ping <address>`	All	Tests whether another device is reachable, and how long the round trip takes.
`arp -a`	All	Shows the local table mapping IP addresses to the MAC addresses behind them.
`tracert / traceroute`	Windows / Unix	Lists every router a packet passes through on its way to the destination.

Diagnostic order: check your own address first, then ping your gateway, then ping something outside. The first failure tells you where the fault is.

04

Types of Connection

Before we ask what shape a network is, we ask how any two points are joined

Point-to-point

Multipoint

Point-to-Point and Multipoint

Two ways to attach devices to a link — every topology is built from these

POINT-TO-POINT

A dedicated link between exactly two devices

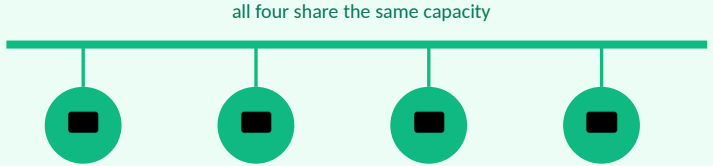


entire capacity reserved for these two

Capacity	The full bandwidth belongs to those two devices
Addressing	Simple — anything sent can only be for the other end
Cost	High — every pair needs its own dedicated link
Examples	PC to switch port · router to router WAN link · a /30 subnet

MULTIPOINT (MULTIDROP)

One link shared by three or more devices



Capacity	Shared — divided among however many are active
Addressing	Essential — each frame must name its destination
Cost	Low — one cable serves many devices
Examples	Old bus Ethernet · Wi-Fi · a hub · cable broadband

The trade-off in one line: point-to-point buys guaranteed performance with cable and ports. Multipoint buys cheapness with contention.

05

Physical Topology

The shape of the network — and why the shape decides cost, resilience, and growth

Bus | Star | Ring | Mesh | Tree | Hybrid

What Do We Mean by Topology?

The arrangement of devices and the links between them

Physical Topology

How the cables actually run. What you would see if you lifted the floor tiles and followed every wire.

Logical Topology

How the data actually flows. These can differ: classic Token Ring was wired as a star but behaved as a ring.

The three questions we will ask about every topology

Cost

How much cable, how many ports, how much labour to install it?

Fault tolerance

If one cable or one device fails, how much of the network goes down with it?

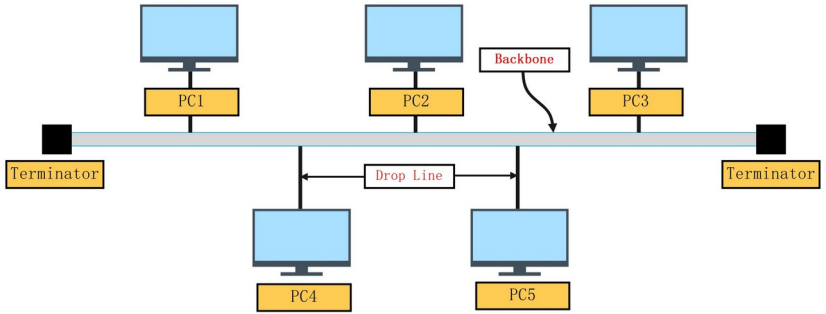
Scalability

How hard is it to add the fiftieth device? And the five hundredth?

No topology wins on all three. Every real network is a compromise, chosen for its budget, its risk tolerance, and its growth plans.

Bus Topology

One backbone cable, every device tapped onto it



How it works

A single backbone cable runs the length of the network. Every device taps into it. A transmission travels the whole cable in both directions and every device sees it, keeping only what is addressed to it. Terminators at both ends absorb the signal so it does not bounce back.

Where you find it

The original 10BASE2 and 10BASE5 Ethernet of the 1980s. Effectively obsolete now, but the concept survives in some industrial and vehicle networks such as CAN bus.

If one link fails

A break anywhere in the backbone splits the network in two and usually stops all communication.

ADVANTAGES

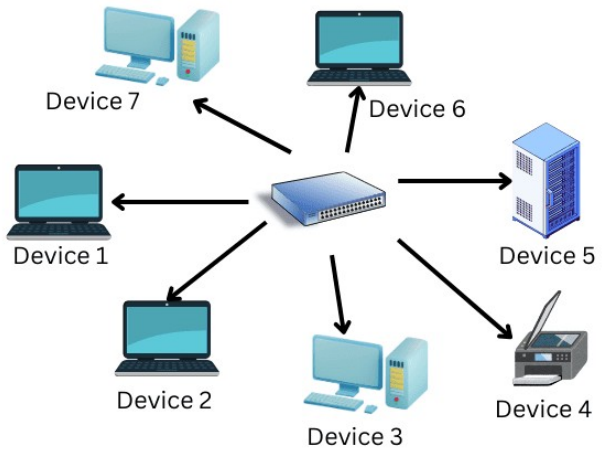
- Uses the least cable of any topology
- Very cheap and simple to install
- Easy to extend by adding cable length

DISADVANTAGES

- One cable fault takes down everything
- Difficult to locate a fault along the run
- Performance collapses as devices are added

Star Topology

Every device gets its own cable to a central switch



How it works

Every device has its own dedicated cable running back to a central switch. Nothing connects directly to anything else. All traffic passes through the centre, which reads the destination address and forwards each frame only where it needs to go.

Where you find it

Essentially every modern wired LAN, including this computer lab, every office, and every home router with Ethernet ports. It is the default choice today.

If one link fails

Only the one device on that cable loses connectivity. Everyone else is completely unaffected.

ADVANTAGES

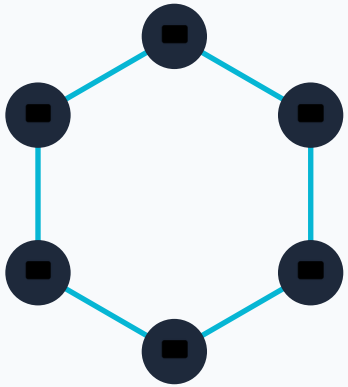
- A single cable fault isolates one device only
- Very easy to add or remove devices
- Faults are simple to locate and fix
- Full bandwidth per device via a switch

DISADVANTAGES

- Needs far more cable than a bus
- If the central switch fails, all devices lose service
- Cost of the central device and its ports

Ring Topology

Each device connects to exactly two neighbours, forming a closed loop



Data flows one way around the loop

How it works

Devices are wired in a closed loop, each joined to the two beside it. Data travels in one direction, passing through every device in turn; each one regenerates the signal and passes it along. Classic ring networks used a token — only the holder could transmit, so collisions were impossible.

Where you find it

IBM Token Ring and FDDI in the 1980s and 1990s. Today the idea survives in some fiber backbones and metropolitan networks, usually as a dual counter-rotating ring for resilience.

If one link fails

In a single ring, one break stops everything. A dual ring can heal itself by reversing direction around the fault.

ADVANTAGES

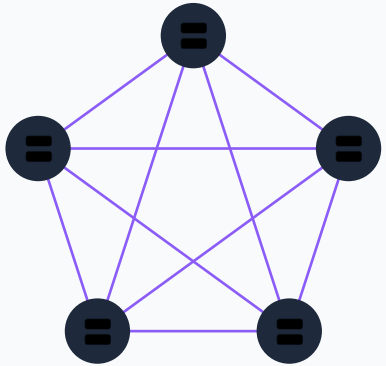
- No collisions — orderly, predictable access
- Performance stays steady under heavy load
- Every device regenerates the signal

DISADVANTAGES

- One break can halt the entire ring
- Adding a device means breaking the loop
- Traffic passes through many devices, adding delay

Mesh Topology

Every device connects directly to every other device



5 nodes need 10 links

How it works

Every node has a dedicated point-to-point link to every other node. With n devices you need n(n−1)/2 links: 5 nodes need 10 cables, 10 nodes need 45, and 50 nodes would need 1,225. A partial mesh connects only the most critical pairs, which is what real networks use.

Where you find it

The Internet backbone between major routers, data centre cores, and any situation where an outage is unacceptable. Also wireless mesh systems in large homes and campuses.

If one link fails

Nothing noticeable. Traffic simply takes one of the many alternative paths available.

ADVANTAGES

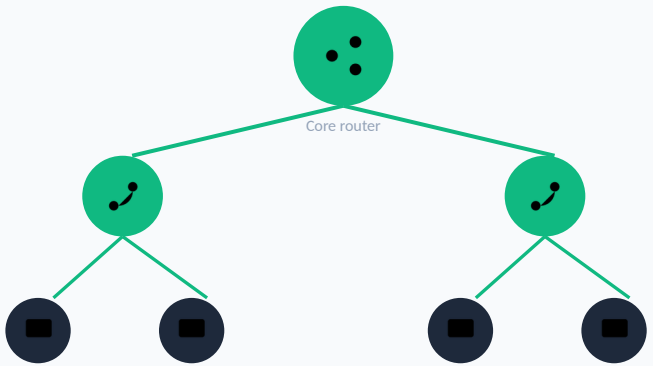
- The most fault tolerant topology there is
- Dedicated links mean no contention at all
- Traffic is private between each pair
- Faults are easy to isolate

DISADVANTAGES

- Enormously expensive in cable and ports
- Complex to install and document
- Impractical beyond a modest number of nodes

Tree Topology

Stars connected to stars — a hierarchy with a root



Hierarchical: core → distribution → access

How it works

A tree is a hierarchy of stars. A root device sits at the top, connected to several intermediate switches, each of which serves a group of end devices below it. Networking professionals usually describe these three levels as the core, distribution, and access layers.

Where you find it

Almost every large organisation. This university's network is a tree: a core router, a switch per building, and access switches on each floor serving the rooms.

If one link fails

Only the branch below the failure is affected. A failed access switch takes out one room; a failed core takes out everything.

ADVANTAGES

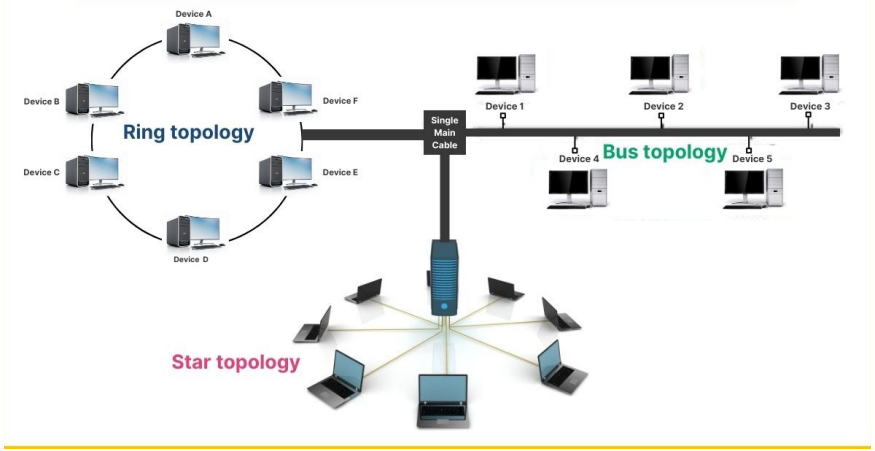
- Scales cleanly to very large networks
- Faults are contained within one branch
- Easy to manage and document by area
- Matches how buildings are actually laid out

DISADVANTAGES

- The root is a critical single point of failure
- More cable and devices than a flat star
- Deep hierarchies add latency between branches

Hybrid Topology

Two or more topologies combined — which is what every real network is



How it works

A hybrid combines topologies so each part of the network uses whatever suits it. A very common pattern is a mesh or partial-mesh core for resilience, feeding tree and star layers for the cheap, easy connection of large numbers of end devices.

Where you find it

Every network of any size. The Internet itself is the largest hybrid there is: a mesh backbone, trees inside organisations, and stars at the edge where your PC plugs in.

If one link fails

It depends entirely which part failed — which is the point. Resilience is bought only where it is needed.

ADVANTAGES

- Each section uses the best fit for its job
- Money is spent on resilience only where it matters
- Very flexible and scalable
- Reflects how organisations really grow

DISADVANTAGES

- Most complex to design and document
- Requires skilled staff to maintain
- Higher initial planning and equipment cost

Comparing All Six Topologies

Cost, fault tolerance, and scalability — the three questions from earlier

Topology	Cost	Fault tolerance	Scalability	Defining characteristic
Bus	Very low	Very poor	Poor	One backbone, tapped
Star	Moderate	Good	Good	All cables to one switch
Ring	Moderate	Poor (single ring)	Moderate	Closed loop, one direction
Mesh	Very high	Excellent	Very poor	Everything to everything
Tree	High	Good per branch	Excellent	Hierarchy of stars
Hybrid	Varies	Designed in	Excellent	Mix, chosen per section

If you remember one row: star is the default for access, tree is how you scale it across a campus, and mesh is what you pay for where downtime is unacceptable.

Choosing a Topology in Practice

Four scenarios — what would you actually build?

A home with 6 devices

Star

One router with a built-in switch and access point. Cheap, simple, and a single fault affects one device.

A university campus

Tree (hybrid)

Core router, a switch per building, access switches per floor. Possibly a mesh between buildings for resilience.

A bank's data centre

Mesh

Downtime costs more than cable. Full or partial mesh between core devices so no single failure is visible.

Sensors in a factory

Bus-like

Industrial buses such as CAN still suit long runs of simple devices where cable cost dominates.

The question to ask: what does an hour of downtime cost this organisation? The answer sets the budget, and the budget picks the topology.

06

Practical Activity

Build and address real topologies in Cisco Packet Tracer

- Star topology
- Addressing a subnet
- Two networks + a router
- Compare hub vs switch

Task 1

Build a Star Topology and Address It

Four PCs, one switch, one subnet

1. Drag four PCs and one 2960 switch onto the workspace.
2. Use Copper Straight-Through to connect each PC to a different switch port.
3. Give the PCs 192.168.10.1 through 192.168.10.4, all with mask 255.255.255.0.
4. Leave the default gateway blank — all four are on the same network.
5. From PC0, ping each of the other three addresses in turn.
6. Open the switch CLI and run show mac address-table to see what it learned.

Success looks like

All three pings succeed, and the MAC table lists four entries.

If it goes wrong

An empty MAC table means no traffic has flowed yet. Ping first, then look again.

Task 2

Two Networks Joined by a Router

Where the default gateway finally starts to matter

1. Build a second star: two more PCs on a second switch.
2. Add a 1941 router and connect each switch to a different router interface.
3. Network A: PCs on 192.168.10.x/24, gateway 192.168.10.254.
4. Network B: PCs on 192.168.20.x/24, gateway 192.168.20.254.
5. In the router CLI, give each interface its gateway address and issue no shutdown.
6. Ping from a PC in network A to a PC in network B.

Success looks like

The cross-network ping replies, and both interface lights are green.

If it goes wrong

Ping fails but the same-network ping works? The gateway on the PC is missing or the interface is still shut down.

Task 3

Hub vs Switch, Seen in Simulation

Prove to yourself what we claimed on the comparison slide

1. Build two identical four-PC networks: one on a hub, one on a switch.
2. Address each network in its own subnet so they stay independent.
3. Switch to Simulation mode and filter to show only ICMP.
4. Send a Simple PDU from PC0 to PC2 on the hub network. Watch every PC receive it.
5. Repeat on the switch network. Watch only PC2 receive it.
6. Write one sentence in your notes explaining what you just saw.

Success looks like

You can state clearly why the hub floods and the switch does not.

If it goes wrong

On the very first switch transmission you may still see flooding. Send a second PDU and it will be unicast.

Summary — Key Takeaways

Everything we covered in Week 2

Hardware	NIC connects a device; repeater fights distance; hub floods; switch forwards by MAC; router forwards by IP; modem converts signals; access point adds radio.
Software	Drivers speak to one specific card. Protocol stacks encapsulate data layer by layer. Network OS services provide files, printing, DHCP, and DNS.
IP addressing	32 bits in four octets. The mask splits network from host. Private ranges are reused everywhere; DHCP hands addresses out; subnetting divides a block.
Connections	Point-to-point gives two devices a dedicated link. Multipoint shares one link among many, so addressing and taking turns become essential.
Topologies	Bus is cheap and fragile; star is today's default; ring is orderly but brittle; mesh is resilient and costly; tree scales; hybrid is what everyone actually builds.
The trade-off	Cost, fault tolerance, and scalability pull against each other. There is no best topology, only the right one for a given budget and risk.

Coming Up in Week 3

Network Types and Classification

This week we studied the devices and the shapes they are wired into. Next week we step back and classify whole networks by their size, ownership, and purpose — and look at what changes when a network spans a city rather than a room.

Networks by Scale

PAN, LAN, MAN, and WAN — what genuinely changes as distance grows.

Client-Server and P2P

Two models of organisation, revisited properly with their real trade-offs.

Intranet, Extranet, Internet

Classification by who is allowed in, rather than by size.

Modern Variants

VPNs, virtual networks, and where the cloud fits into all of this.

To prepare

Practise

Convert 172, 200, and 240 into binary without looking at the slide

Rebuild

Today's two-network router topology, from memory

Look ahead

Assignment 1 is issued next week and uses all of this

Thank You

Any Questions?